Smart Cities: AI-Based Vehicle Tracking and License Plate Recognition System

PROBLEM STATEMENT:

Modern smart cities face increasing traffic congestion, law violations, and inefficient monitoring systems.

Manual surveillance is costly, slow, and error-prone.

This project aims to build an AI-powered automated traffic monitoring system that can detect vehicles, count them, and recognize license plates in real-time for better traffic management and law enforcement.

OBJECTIVES:

- Detect vehicles (car, bike, truck, bus) in real-time video streams

- Count vehicles crossing a virtual line

- Extract license plate numbers using OCR, if vehicle cross the virtual line

- Store vehicle type, license plate and timestamp in a database

- Provide analytics for smart city traffic management

SYSTEM ARCHITECTURE:

Video Input

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Frame Extraction (OpenCV)

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YOLOv8 Vehicle Detection Model (pretrained on COCO dataset)

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DeepSORT / ByteTrack Tracking

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Line Crossing Detection Module

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License Plate Detection Model (YOLO ANPR)

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OCR Engine (EasyOCR / Tesseract)

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Database Storage (SQLite / PostgreSQL)

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Streamlit Dashboard / FastAPI Service

STEP-BY-STEP WORKFLOW:

1. Data Input

- Load CCTV footage or traffic dataset videos
- Extract frames using OpenCV

2. Vehicle Detection

- Use pretrained YOLOv8 model (COCO dataset)
- Detect:

  - Car
  - Bike
  - Truck
  - Bus

3. Tracking

- Apply DeepSORT or ByteTrack
- Assign unique ID to each vehicle
- Ensure same vehicle is not counted multiple times

4. Line Crossing

- Define a virtual line on road
- Detect centroid crossing event
- Increment vehicle counter per class

5. License Plate Detection

- Apply YOLO-based license plate detection model
- Trigger only when vehicle crosses line

6. OCR

- Crop license plate region
- Apply EasyOCR / Tesseract
- Convert image → text (plate number)

7. Database (SQLite)

Store structured records:

  - Vehicle type
  - License plate number
  - Timestamp of event

8. Dashboard

- Live vehicle count (class-wise)
- License plate logs
- Traffic analytics graphs
- Historical reports

DATASET LINK (License plate dataset):

https://universe.roboflow.com/roboflow-universe-projects/license-plate-recognition-rxg4e

EXPECTED OUTPUT:

- Vehicle detection and counting

- License plate recognition

- Traffic analytics dashboard

- Stored logs in database

Database Schema (Structure of table)

- Vehicle Type
- Plate Number
- Time Stamp

Sample Vehicle Log Data

☞ "10 trucks crossed the line"

Database will store:

Vehicle Type	Plate Number	Timestamp
Truck	ABC-123	10:01 AM
Truck	XYZ-456	10:02 AM
Truck	DEF-789	10:05 AM

Sample Videos for Testing (Links)

https://drive.google.com/drive/folders/11n5u1B3BAppISJl7OgLNII3shY9FnLsV?usp=sharing

Folder structure:

```
smart-traffic-anpr/
│
├── streamlit_app.py
├── data/
├── notebooks/
├── models/
├── database/
├── output/
├── requirements.txt
├── README.md
```

Deliverables:

Github Repository link

Streamlit App (Deployed)

TECH STACK:

Python, OpenCV, YOLOv8, DeepSORT, EasyOCR, FastAPI, Streamlit

IMPACT:

This system helps smart cities automate traffic monitoring, reduce manual surveillance, and improve law enforcement efficiency.